Rounding to Significant Figures

Exam code: 0607

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Rounding to significant figures

How do I round a number to a given significant figure?

- To find the **first significant figure** when reading from left to right, find the **biggest place value** that has a **non-zero digit**
  - The first significant figure of 3097 is 3
  - The first significant figure of 0.006207 is 6
    - The zeros **before** the 6 are **not** significant
    - The zero **after** the 6 **is** significant
- Count along to the **right** from the first significant figure to **identify** the **position** of the **required significant figure**
  - **Do** count zeros that are between other non-zero digits
    - E.g. 0 **is** the **second** significant figure of 3097
    - 9 is the **third** significant figure of 3097
- Use the **normal rules for rounding**
  - Count in **units** determined by the **place value** of the significant figure
    - If the second significant figure is in the 10's column, count in 10's
  - **Identify** numbers it could be rounded **up** or **down** to
  - **Circle** the number to the **right** of the significant figure
  - Use this value to determine which number it rounds to
- For **large numbers**, complete places **up to the decimal point** with **zeros**
  - E.g. 34 568 to 2 significant figures is 35 **000**
- For **decimals**, complete places **between** the **decimal point** and the **first significant figure** with **zeros**
  - E.g. 0.003 435 to 3 significant figures is **0.00**3 44

How do I know what degree of accuracy to give my answer to?

- If a question requires your answer to be an **exact value**
  - You can leave it as a **simplified fraction**

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

        - E.g. 4π, or $\sqrt{3}$
    - If it is an **exact decimal** up to and including 5 s.f., you can write it out without rounding it
        - E.g. 0.9375, or 850.25

- If the answer is **not exact**, an exam question will often state the **required degree of accuracy** for an answer
    - E.g. Give your answer to 2 significant figures

- If the degree of accuracy is not asked for, use **3 significant figures**
    - All **working** and the **final answer** should show values correct to at least 4 significant figures
    - The final answer should then be **rounded** to 3 significant figures

- In **money calculations**, unless the required degree of accuracy is stated in the question, you can look at the **context**
    - Round to **2 decimal places**
        - E.g. £64.749214 will round to £64.75
    - Or to the nearest whole number, if this seems sensible (for example, other values are whole numbers)
        - £246 029.8567 rounds to £246 030

- When calculating **angles**, all values should be given correctly to **1 decimal place**
    - An angle of 43.5789° will round to 43.6°
    - An angle of 135.211...° will round to 135.2°

When might I need to estimate up or down?

- If a **real life scenario** is given, this may influence if you need to round **up or down**
- For example, if a class of 31 students requires at least 1 adult for every 10 children
    - 31 ÷ 10 = **3.1** adults are needed
    - 3 adults would not be enough, this would only be enough for 30 students
    - Therefore you would **round up**, to **4** adults

- Alternatively, consider a farmer filling crates that each hold 12 apples
    - How many crates can he fill if he has 50 apples?
        - 50 ÷ 12 = 4.16... crates will be filled
        - He cannot make a 5th full crate, as this would require 60 apples
        - Therefore, you would round down, to 4 crates

Examiner Tips and Tricks

- In an exam question check that you have written your answer correctly by considering if the value you have ended up with makes sense.

(without the zeros) and 2 530 000 are very different sizes!

Worked Example

Round the following numbers to 3 significant figures.

(i) 345 256

(ii) 0.002 956 314

(iii) 3.997

Answer:

(i)

The first (non-zero) significant digit is in the hundred thousands column (3)
The third significant figure is therefore the value in the thousands column (5)

Count in thousands
Identify the numbers that it could round down to (345 000) or round up to (346 000)

Circle the digit on the right of the third significant figure (2)

$$345\,2\,56$$

This digit is less than 5 so round down

345 000 (3 s.f.)

(ii)

The first significant digit is in the thousandths column (2)
The third significant figure is therefore in the hundred thousandths column (5)

Count in hundred thousandths
Identify the number it could round down to (0.00295) or round up to (0.00296)

Circle the digit to the right of the third significant figure (6)

$$0.002\,95\,6\,314$$

6 is greater than 5 so we need to round up

0.002 96 (3 s.f.)

(iii)

The first significant digit is in the units column (3)
The third significant figure is therefore in the hundredths column (9)

Count in hundredths
Identify the number it could round down to (3.99) or round up to (4.00)

Circle the digit to the right of the third significant figure (7)

3.997

This value is greater than 5 so it will round up

- The two zeros are necessary to indicate that it has been rounded to 3 significant figures

4.00 (3 s.f.)

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